Benchmark Testing the Flow and Solidification Modeling of Al Castings

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Although the heat flow aspects of the simulation of castings now appears to be tolerably well advanced, a recent exercise has revealed that computed predictions can, in fact, be widely different from experimentally observed values. The modeling of flow, where turbulence is properly taken into account, appears to be good in its macroscopic ability. However, better resolution and the possible general incorporation of surface tension will be required to simulate the damaging effect of air entrainment common in most metal castings. It is envisaged that the results of this excercise will constitute a useful benchmark test for computer models of flow and solidification for the foreseeable future.

INTRODUCTION

As part of 1995's 7th Conference on Modeling of Casting, Welding and Advanced Solidification Processes and with the goal of developing a well-characterized casting experiment that would be a useful benchmark for the modeling community, a team at the University of Birmingham, United Kingdom, devised a simple plate design for group testing. The geometry had to be as simple as possible since the metal and heat flow aspects of the models were to be tested—not the capabilities of computer-aided design packages. However, the filling system was less easy to model because of its delicate geometry. A short sprue, common for many small castings, would have been relatively easy to model because of the modest velocities. A tall sprue was chosen, however, since such a sprue would give a high degree of turbulence in the runner and gate, and this would represent an interesting test of the abilities of current computer models.

Participants who took part in this comparison exercise[1-9] succeeded in creating truly impressive simulations. They demonstrated how far computer simulation has come in the last few years. More detail of their models may be found in the proceedings of the conference.[10]

Table I. Mold Dimensions (mm)

Axis	x	y	z
Sprue Entrance	15	19.2	—
Sprue Exit	15	5	—
Sprue Height	—	—	380
Well	15	5	10
Runner	15	240	20
Gate	10	60	15
Plate	10	200	100

EXPERIMENTAL DETAILS

The overall geometry of the casting—a simple plate with a simple bottom-gated running system (and no feeder)—is depicted by Figure 1 and Table I. The mold was made from 60 AFS-grade washed-and-dried silica sand, bonded with 1.2 wt.% phenolic urethane resin (Ashland Pepset). This is a widely used molding material, and it was hoped that good material properties would be available. The choice of a sand mold assisted the viability of analysis by x-ray radiography and simplified computation to some extent because its high permeability to air would reduce the back pressure to a minimum due to entrapped mold gas. The liquid metal used was 99.999% aluminum; again, this choice was made in order to obtain unambiguous and accurate material data.

A metal charge of 2.2 kg was poured at approximately 720°C into the pouring basin. The basin was full (signaled by the escape of liquid into an overflow basin) when filled to a depth of precisely 40 mm. At this instant, the stopper was abruptly lifted clear of the basin, beginning the filling of the sprue. The temperature at which metal entered the sprue was approximately 700°C. The pouring continued uninterrupted for 3.5 s.

A weir in the pouring basin ensured that only negligible bulk turbulence in the liquid was present above the entrance to the sprue. The stopper had a tapering square section to allow a close fit into the entrance of the square section sprue.

X-Ray Radiography

Mold filling was observed with a 160 kV x-ray source (1.5 mm diameter) equipped with an image intensifier. Recording was performed using a VHS 50 Hz recorder.

Table II. Experimental Freezing Time Measurements at the Thermocouple Locations

	Solidification Time (s)	
Location	Melt 1	Melt 2
A	63.00	63.67
B	56.95	60.20
C	19.75	18.55
D	13.10	13.20
E	12.75	10.63
F	15.35	15.27
G	(22.0)	11.94

Pouring in the lead-lined cabinet was carried out remotely, outside the cabinet, by an operator viewing the pour via a closed-circuit television camera.

Unfortunately, because of the limited field of view, not all of the mold could be viewed at one time by the video unit. The start of the pour in the region of the pouring basin is seen in Figure 2. In strict terms, this was not a required part of the benchmark test, although some participants requested details and provided a model of the basin.

The instant when the stopper was lifted was designated as time zero. Subsequently, x-ray video frames were taken at 0.02 s intervals during filling. The experiment was repeated several times. In Figure 3, three sequences are shown. The figure presents video excerpts at 0.25 s intervals during the filling of three separate castings to ascertain the degree of reproducibility of the pour. The researchers were careful to ensure that the experimental conditions in each case were reproduced as exactly as possible. The only difference in the sequences is that the geometry of the view was changed slightly so as to see the sprue on one occasion and the ingate area and end of the runner on the other occasions.

In print terms, the reproduction quality of these images is limited compared to the quality of the original video recordings. Many interesting features of the flow, particularly the large number of bubbles that were entrained from time to time, are visible only at very much reduced contrast. Comparison of the three sequences indicates interesting similarities and differences, which underscore the difficulties of carrying out experiments designed to test numerical models and the limits as to what degree of agreement can be expected.

Table III. A Comparison of Calculated vs. Experimental Results

		Solidification Time (s)		
Ref.	Fill Time (s)	Shortest	Longest	Ratio
Exp.	2.0	11.7	63.3	5.41
1	2.15	20	50	2.50
2	2.4	—		—
3	2.06	13	53	4.08
4	—	—	—	—
5	2.20	50	92	1.84
6	1.83	5	35	7.00
7	2.3	45	65	1.44
8	2.2	25	80	3.20

Thermal Analysis

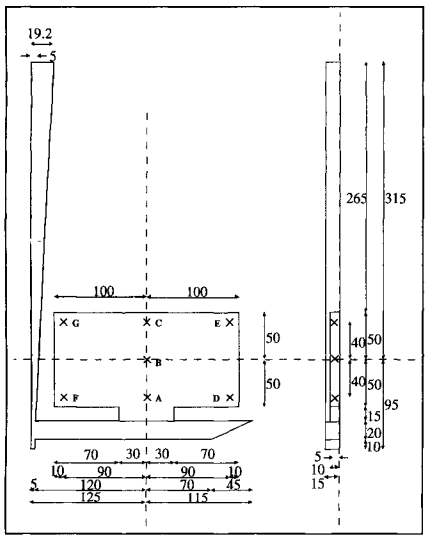
Using thermocouple wires of 0.3 mm diameter, a junction was created by crossing the wires and spot welding, thus avoiding the formation of a bead. The excess wire was trimmed off. The wires protruding into the mold cavity were insulated with white paint (TiO_2-containing emulsion paint) up to 1 mm of the junction, which was left unprotected. The paint was then fired to burn off volatiles. Care was taken to ensure that wires could not short circuit at the mold wall (a common source of error, which gives an effective temperature reading corresponding to a position close to the metal/mold interface). In any case, a certain amount of error is to be expected from the readings since it was noted that shrinkage cavities formed around the bead of the thermocouple. This will, of course, reduce heat flow and local mass of metal as well as the consequent amount of local latent heat that will be evolved. The net effect is, therefore, expected to be to shorten somewhat the overall length of freezing-time measurement.

Readings were data logged into a computer at a rate of 50 per second. Experience in the University of Birmingham indicates that such a geometry of thermocouple has a rate of response that will easily follow the thermal environment in 15 mm thick aluminum plates. Again, pains were taken to carry out two repeat experiments as identically as possible.

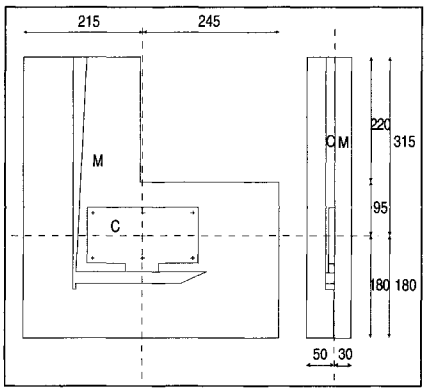
Materials Data

The materials data for pure aluminum were taken from *Metals Handbook*, Rolls Royce, Magma Gmbh, and others. The data were used by several workers. However, several values were questioned by some participants. These were a wide freezing range, which indicated that the data applied more accurately to commercially pure aluminum (98%) than the pure material intended for the benchmark experiment. Further confusion surrounded the viscosity of liquid aluminum, which was quoted as a number (4×10^{-6}) without units. Units were then added (Pa · s), unfortunately in error, based on the belief that viscosity was dynamic viscosity. In fact, the parameter was kinematic viscosity and should have been designated as m^2s^{-1}. *Smithells Metals Reference Book* gives the dynamic viscosity of pure aluminum at the freezing point as 1.3×10^{-3} Pa · s and its density as 2.385×10^3 kgm^{-3}.[11] This yields a kinematic viscosity of 0.55×10^{-6} m^2s^{-1}.

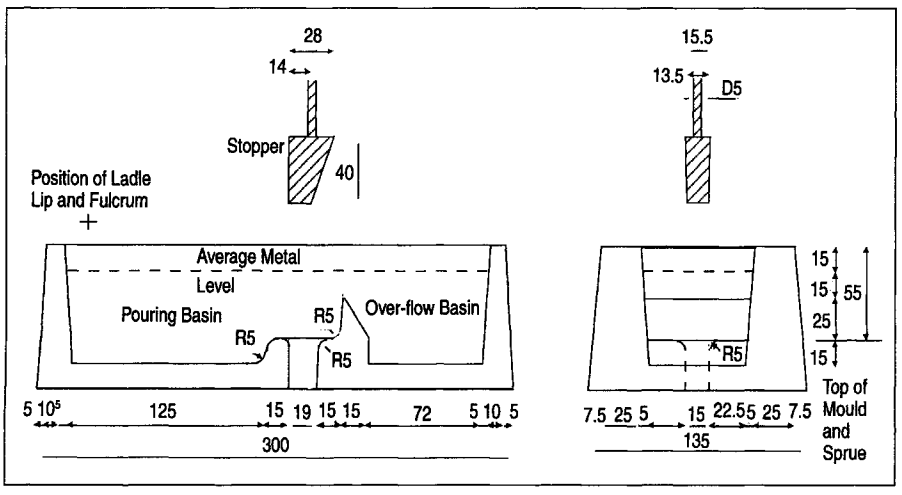
Agreement on a database was the original aim of the exercise, and it would have greatly assisted in comparing re-



a



b

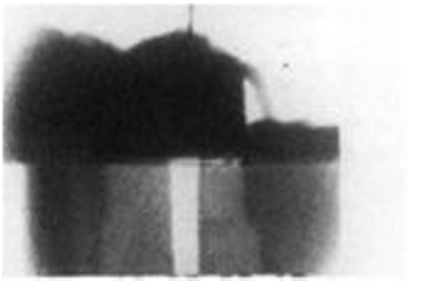


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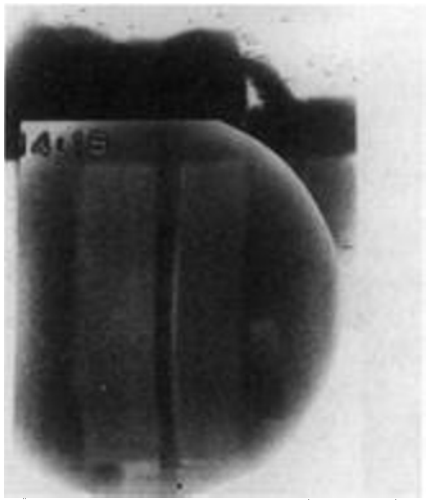
Figure 1. Benchmark test dimensions: (a) casting dimensions (mm) and thermocouple locations (denoted by an "x"); (b) mold dimensions (mm); and (c) pouring basin and stopper dimensions (mm).



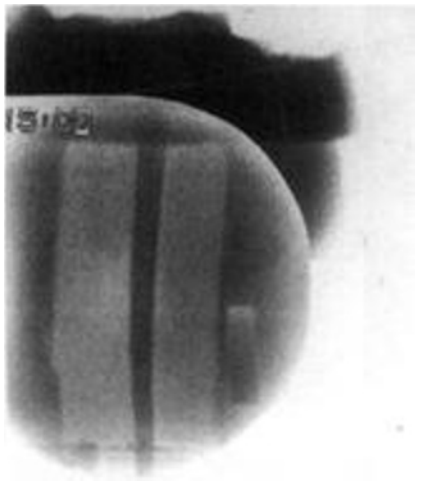
a



b



c



d

Figure 2. Real-time x-ray radiographs of the filling of the pouring basin and sprue: (a) condition prior to pour and stopper removal, (b) stopper removal, (c) 0.24 s after stopper removal, and (d) 0.5 s after removal.

sults. It has to be admitted that this aspect of the comparison has not worked out as well as was intended. The differences in data from various sources underscores a problem that continues to hinder all efforts to simulate castings.

RESULTS

Heat Transfer

The measured cooling curves are presented in Figure 4. The agreement between the thermal analysis traces from two castings appears to be reasonable, although significant differences can be noted. The major differences relate to freezing times (Table II). Although most data agree reasonably well, site G shows a difference of nearly a factor of two. Close inspection reveals that it is melt number one that is in serious error. This value is, therefore, shown in brackets.

Again, these results highlight the problem of carrying out such experiments, even when conducted by a lab and experimenters of considerable experience using the best current practice. It is clear that differences between experimental data and numerically computed results cannot always be assumed to be the problem of the numerical simulation.

Nevertheless, the predicted freezing times from the first volunteer modelers displayed a wide scatter of disagreement (Table III). Where freezing time data were not tabulated by participants, approximate values were determined from their graphical results. To identify whether disagreements were simply the result of using different thermophysical data or, more fundamentally, differences in models, the longest and shortest predicted freezing times in the plates were compared. If only thermophysical data were involved, then both long and short predicted times would be expected to be increased or decreased at least approximately in proportion to the experimentally measured times. For some models, this was approximately true; in other models, however, there was considerable disagreement.

Nevertheless, it was reassuring to note that all of the simulations correctly predicted (or very closely predicted) that the last part of the casting to freeze was that nearest to the ingate and not that at the geometric center of the plate, which illustrated a useful degree of internal consistency within each of the models.

Flow

Some of the most interesting differences between the models related to the predictions of the flow of the liquid metal. Figure 5 illustrates the character of three main classes of predictions.

- Figure 5a assumes highly viscous laminar flow, in the manner of many polymers.
- Figure 5b shows essentially laminar flow, in which the fast underlying flow along the bottom of the runner does not mix with the laminar flow of the reflected wave that travels above. Thus, the momentum of the stream impacting on the side of the gate nearest to the sprue carries the liquid to fountain to the left. (Although such a fountain to the left has previously been assumed to result from entrapped air escaping from the runner, the high permeability of the chemically bonded sand mold used in this work would be expected to give rise to negligible back pressure. The effect might be important in impermeable metal dies.)
- Figure 5c shows a model in which turbulent mixing between streams is assumed to be complete, so that the reflected wave is now compressed into the end of the runner. Thus, metal in the end of the runner experiences momentum only from the direction of the sprue. Therefore, as soon as the reflected wave reaches the gate, it expands rapidly into the gate, and, with momentum from the direction of the sprue, fountains to the right.

Some models fall between these three extreme illustrations. However, it is clear that the case model depicted in Figure 5c, assuming a high degree of turbulent

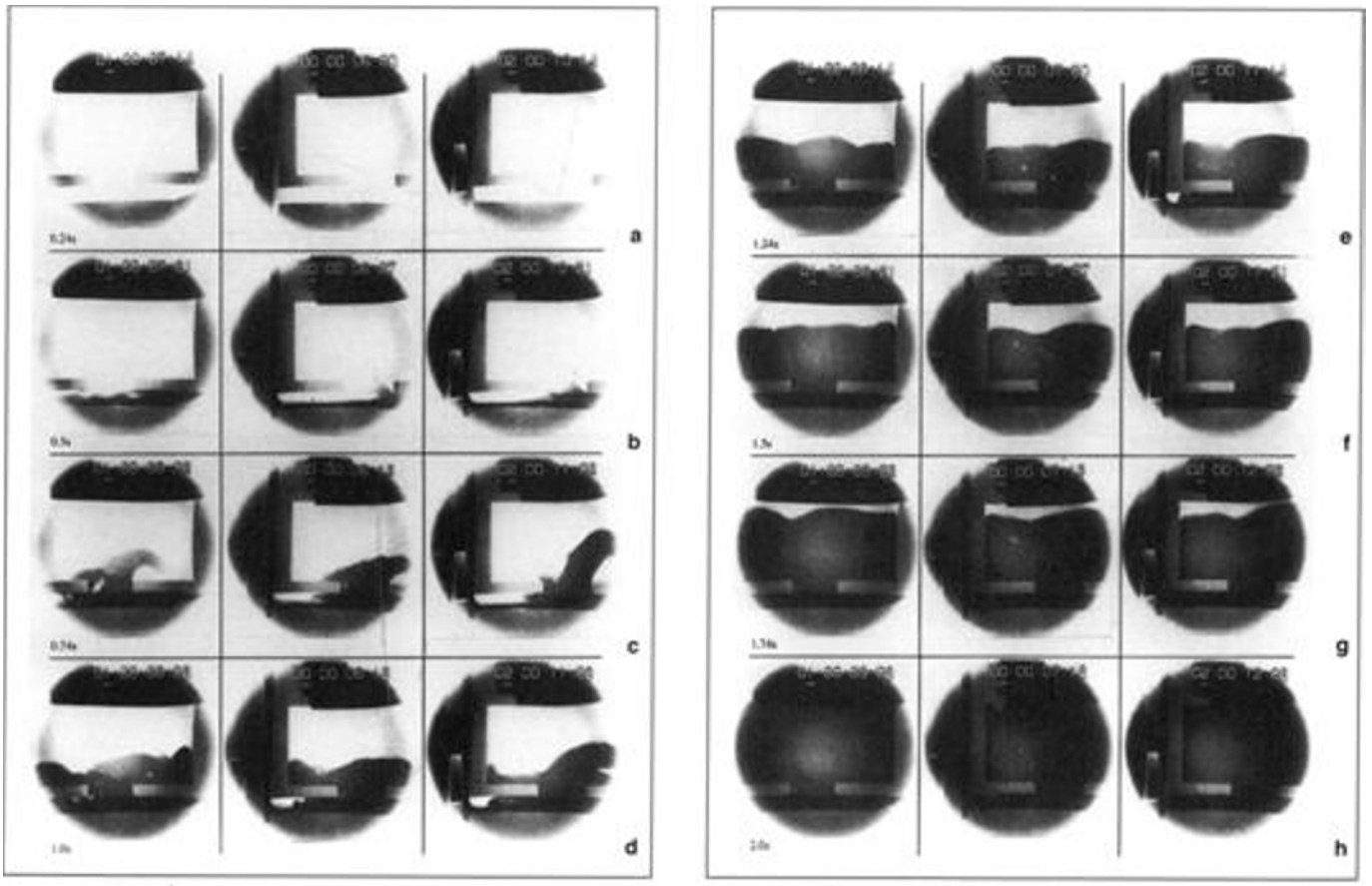


Figure 3. Three experiments run on the real-time x-ray unit to compare the progressive filling of the mold: (a) 0.24 s after stopper removal, (b) 0.5 s, (c) 0.74 s, (d) 1.0 s, (e) 1.24 s, (f) 1.5 s, (g) 1.74 s, and (h) 2.0 s.

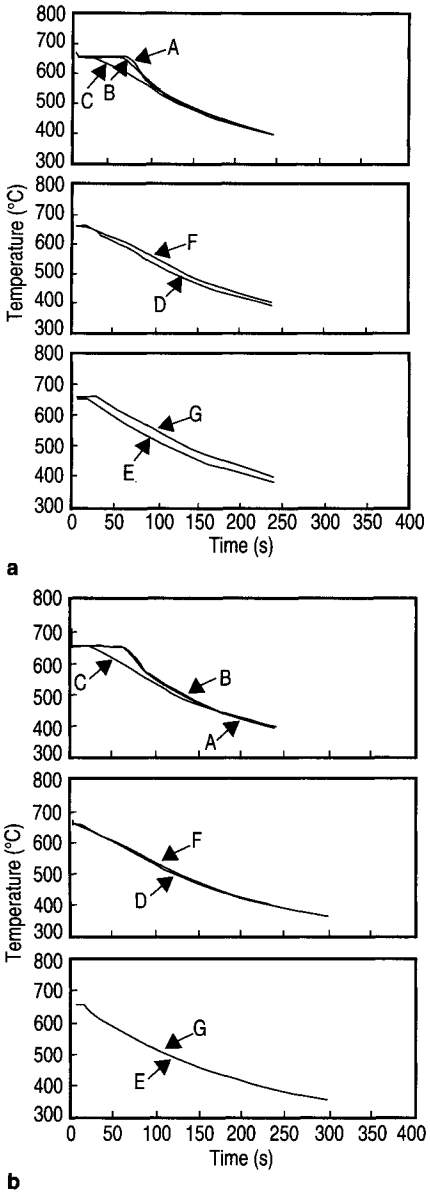


Figure 4. A consistency check for the thermal analysis results at seven thermocouple locations (Table II) in the mold for two melts: (a) melt number one and (b) melt number two.

mixing, gave the nearest approximation to the experimentally observed results. This was consistent with high Reynolds numbers (40,000 or more).

It should be noted that slight differences in behavior can be seen in experiments, even though they were designed to repeat identically. This is to be expected and is in the nature of turbulent phenomena.

Most models predict the filling time correctly at around 2.0 seconds (Table III). (It should be noted that although this time was given, it may have been used as an adjustable parameter by some workers—the time for filling was originally quoted to participants, erroneously, at 2.2 seconds.) Most models correctly predict the fast jet flow along the base of the runner during the early stages of filling the runner. This is an important aspect of the flow along a runner of this depth. Most also correctly predict the

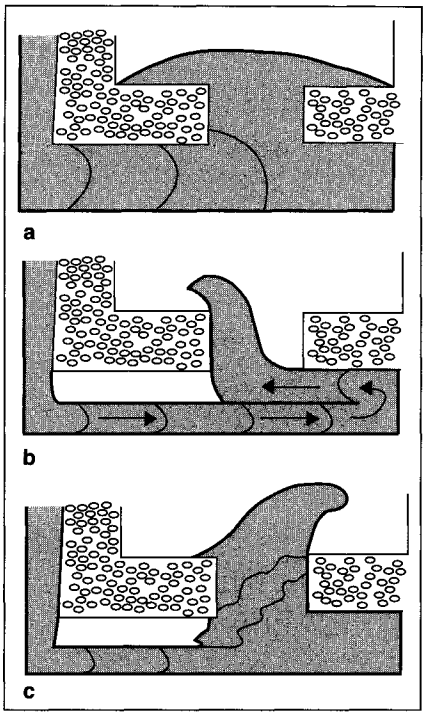


Figure 5. The primary flow patterns predicted by the computer models: (a) highly viscous laminar flow, (b) low-viscosity laminar flow, and (c) turbulent flow. The latter pattern most accurately fits the experimental results.

filling of the runner from its far end as the reflected wave progresses backward. This is also a key feature of flow in this part of the running system for a runner of this depth.

It is of interest that none of the models were able to predict the details of bubble formation, which characterized the junction between the overlying reflected stream and the underlying jet from the base of the sprue. As filling progressed, this partly filled runner phenomenon later evolved into a phenomenon analogous to a hydraulic jump; later still, it evolved into a simple vena contracta, which itself slowly contracted and disappeared at about 1.24 seconds. Some models gave poor predictions of the time of this disappearance.

In general, of course, most models were a long way from having the resolution to predict the bubble formation events, particularly at the hydraulic jump. Both Lewis[2] and Rigaut[3] correctly draw attention to this problem.

It is interesting that the only model to incorporate surface tension was that by Xu and Mampaey[5] (although others claim to be able to incorporate this when necessary[6]). The incorporation of surface tension has thus far not been shown to be an important influence on the predictions in this exercise. However, it would seem to be a useful starting point for the detailed examination of the important and destructive phenomenon of bubble entrainment, which is so widespread in poorly designed filling systems for castings. It seems likely that a large proportion of defects in castings are the result of bubble damage.[12]

Finally, those modelers who also calculated the velocity vectors in the liquid during the filling of the mold cavity mostly noticed the two vortices on either side of the main filling stream. The looped trajectory of bubbles observed on the x-ray video in this area of the mold has confirmed that there are indeed such vortices. Furthermore, this circular trajectory of sequential bubbles has been so closely similar that its occurrence has suggested that the second bubble may be traveling in the oxide trail of the first bubble.[13]

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